

A resonance phase locates unstable self-similar singularities beyond an exponential precision wall

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Abstract

Unstable self-similar blow-up profiles of incompressible flow come in hierarchies, but their computable depth is unknown. Here we show that direct computation meets an exponential precision wall: each profile is a zero of a smoothness defect that shrinks about tenfold per profile. In incompressible porous-media flow, auditing every error source reveals an erratic defect floor of 10^{-9} . Two independent solvers agree to that floor, yet their profile positions drift apart by five orders of magnitude, and a pre-registered holdout test could not identify the seventh profile. A resonance phase, a quadrature over $O(1)$ -accurate profile data, moves 10 to over 1,000 times less under the same numerical changes, locates that profile ten times more precisely and predicts the wall through its imaginary part. A tracking failure that passed residual checks was caught by a geometric invariant and repaired with an exact scaling. Three deeper profiles are registered as benchmarks.

Main

Whether the incompressible Euler equations can develop a singularity in finite time from smooth initial data is one of the central open problems of mathematical physics. Near a solid boundary it has been resolved with computer assistance: Luo and Hou observed explosive growth at the wall of a rotating cylinder¹, and Chen and Hou proved stable, nearly self-similar blow-up for the axisymmetric Euler equations and for their two-dimensional proxy, the Boussinesq system with a boundary^{2,3}. Stable blow-up is, however, only the bottom of a hierarchy. Physics-informed neural networks^{4,5} have recently found the first few *unstable* self-similar profiles of the Boussinesq equations, of incompressible porous-media flow (IPM) and of the Córdoba–Córdoba–Fontelos model^{6,7}, and these profiles are now being computed towards machine precision⁸. Unstable profiles are the thresholds between qualitatively different evolutions and the candidates for blow-up wherever the stable scenario is unavailable; computer-assisted proofs need them to high accuracy^{3,9,10}. How deep such a hierarchy can be computed, and with what certainty a computed profile is the profile it is claimed to be, are therefore computational questions with mathematical consequences.

Here we show that these hierarchies carry an intrinsic obstruction, an exponential precision wall, and we give an observable that removes it. The profiles are isolated members of a continuous branch of singular self-similar solutions; each sits where a smoothness defect vanishes, and that defect is exponentially small in the profile number. In IPM it falls about tenfold per profile,

so each new profile demands ten times more precision. We audit the complete error budget of a converged solver and find that, deep in the hierarchy, it is dominated by discretization errors that are neither small nor smooth: a periodic locking of a receding front to the grid, an angular-resolution error that stops converging, and round-off from deep origin truncation. An independent solver confirms that the floor is a property of the problem, and a holdout test, pre-registered with a fixed decision rule, shows its consequence: the seventh IPM profile cannot be identified by its defect.

The remedy is a different observable. The hierarchy is created by a short wave trapped in a stalled boundary layer, and the smooth profiles are resonances of that wave. Its phase is a quadrature over $O(1)$ -accurate profile data; it is insensitive to the errors that dominate the defect, locates the seventh profile ten times more precisely, and its imaginary part predicts the wall itself. We also report a failure of the phase computation that passed every residual check, the geometric invariant that exposed it and the exact scaling that repaired it. Deeper in the branch a steepening front limits the present method, and we leave the endpoint of the IPM hierarchy open while registering three deeper profiles, with the precision a test would need, as a benchmark.

Smooth profiles are zeros of an exponentially small defect

We seek self-similar solutions $\rho(x, t) = (1 - t)^\lambda R(y)$, $y = x/(1 - t)^{1+\lambda}$, of IPM on the half-plane (Methods). For each blow-up rate parameter λ there is a least-singular profile whose density behaves near the stagnation point on the wall as $R \simeq c|y_1|^m$, with

$$m(\lambda) = \frac{\varepsilon}{1 + \lambda - A(\lambda)}, \quad \varepsilon = \lambda, \quad (1)$$

where A is the strain at the stagnation point. The profile is smooth exactly when $m = 2$. We follow this branch continuously in $z = 1/\lambda$ by Newton–Krylov continuation of a marching solver in log-polar coordinates (Methods)^{11,12} and locate the zeros of the defect $m - 2$. Along the branch the self-similar flow on the wall develops a slow layer, a dip and a receding front (Fig. 1a), which we return to below.

Along the branch the defect oscillates with an amplitude that falls geometrically (Fig. 2a). Its extrema are 8.05×10^{-2} , 7.70×10^{-3} , 7.71×10^{-4} , 7.55×10^{-5} , 7.04×10^{-6} , 5.45×10^{-7} and 5.07×10^{-8} , falling by a factor of 10.0–12.9 per half-period. At its zeros, the smooth profiles $\lambda_1, \dots, \lambda_7$, the defect crosses zero with slopes $|dm/dz| = 5.4 \times 10^{-2}$, 5.9×10^{-3} , 6.4×10^{-4} , 6.6×10^{-5} , 6.6×10^{-6} , 4.6×10^{-7} and 2.5×10^{-8} . A zero located from a defect known to $\pm\sigma_m$ is uncertain by $\sigma_z = \sigma_m/|dm/dz|$, so any fixed error floor is overtaken and σ_z grows by a factor of 10–20 per profile (Fig. 2c). The precision needed to locate profile n is set by the profile number, not by the order of the method. Defects that are beyond all orders in a small parameter^{13–15} are the rule for hierarchies generated by a slow layer; the 2D Boussinesq and Hou–Luo hierarchies behave in the same way (Supplementary Note 1).

An independent solver sees the same wall. A global sparse Newton solver that shares no numerical ingredient with the marching solver imposes $m = 2$ directly and treats λ as an eigenvalue (Methods). It reproduces the IPM profiles $n = 1$ –6 to $|\Delta\lambda| = 1.4 \times 10^{-8}$, 3.7×10^{-8} ,

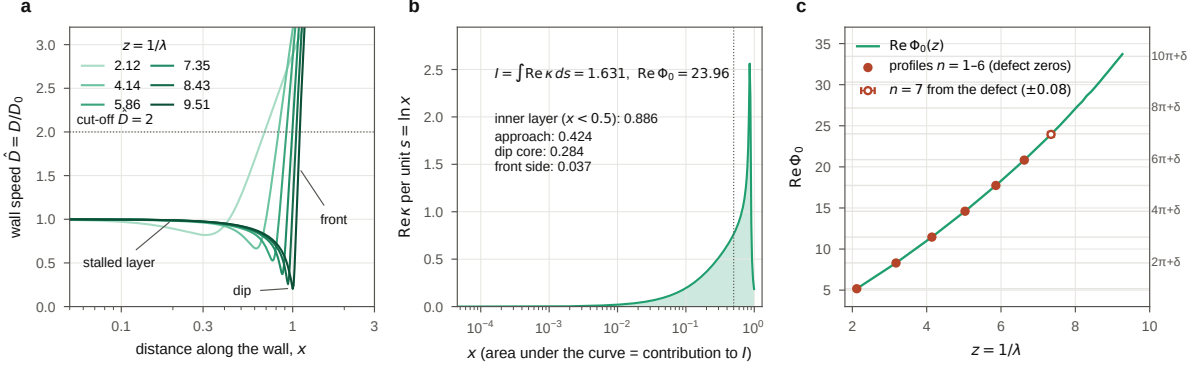


Fig. 1 | The object and the observable (IPM). **a**, Wall speed $\hat{D} = D/D_0$ along the wall for profiles of increasing depth $z = 1/\lambda$: a stalled layer ($\hat{D} \approx 1$), a dip that deepens, and a front that recedes. Dotted line: the front cut-off of the phase. **b**, Integrand $\text{Re } \kappa$ of the resonance phase per unit $s = \ln x$ at $z = 7.346$ (logarithmic x axis, so area is contribution), with the partition of $I = D_0 \text{Re } \Phi_0$ into inner layer, approach, dip core and front side. **c**, $\text{Re } \Phi_0$ along the branch against the levels $n\pi + \delta$ ($\delta = 2.031$). Filled symbols: profiles $n = 1-6$ located as zeros of the defect; open symbol: $n = 7$ with the systematic uncertainty of the defect.

3.4 $\times 10^{-8}$, 6.5 $\times 10^{-7}$, 9 $\times 10^{-7}$ and 1.3 $\times 10^{-4}$. Converted with the measured crossing slopes, every one of these differences corresponds to 0.2–3.5 $\times 10^{-9}$ in m : the two solvers agree to the same floor in the defect at every depth, while their disagreement in profile position grows by five orders of magnitude (Fig. 2c). The wall is a property of the problem, not of a solver. The same pair of solvers reproduces all eight smooth profiles of the 2D Boussinesq system to 10^{-5} (Supplementary Note 1).

An audit of the error floor

We audited every numerical parameter at the deepest profiles (Fig. 3 and Extended Data Table 1). The production settings are angular collocation with $N_b = 32$ points, radial spacing $h_s = 0.0125$ in $s = \ln r$, origin truncation $s_{\text{start}} = -20$, far field at $s_{\text{max}} = 100$ and a Newton tolerance near the residual floor. Three findings matter beyond this problem.

Front-grid locking. As λ decreases, a front on the wall recedes across the radial grid and modulates m with the period of one grid cell in the front's position. We measured the modulation directly by solving the same λ on grids shifted by $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{3}{4}$ of a cell (Supplementary Note 2 and Supplementary Fig. 1). Its half-range is 1.5 $\times 10^{-6}$, 1.2 $\times 10^{-8}$ and 8 $\times 10^{-11}$ at $h_s = 0.025$, 0.0125 and 0.00625, falling as $h_s^{7.1}$ (Fig. 3a). Averaging over the four shifts cancels its first three harmonics. On the coarsest grid it produced a spurious zero of $m - 2$ at $z = 6.905$ and displaced the sixth profile by 0.037 in z , or 5% of a spacing. A plausible-looking but spurious profile is the characteristic failure of direct location.

Angular resolution stops converging. At the seventh profile ($z = 7.346$, $h_s = 0.00625$), $m - 2$ moves by -1.1×10^{-9} from $N_b = 32$ to 48 and by a further -2.3×10^{-9} from 48 to 64, whereas at the sixth profile $N_b = 48 \rightarrow 64$ changes m by only 9×10^{-11} . The floor is erratic, not a convergent tail that extrapolation could remove.

Deeper truncation is not better. Starting the march from the exact local solution at s_{start}

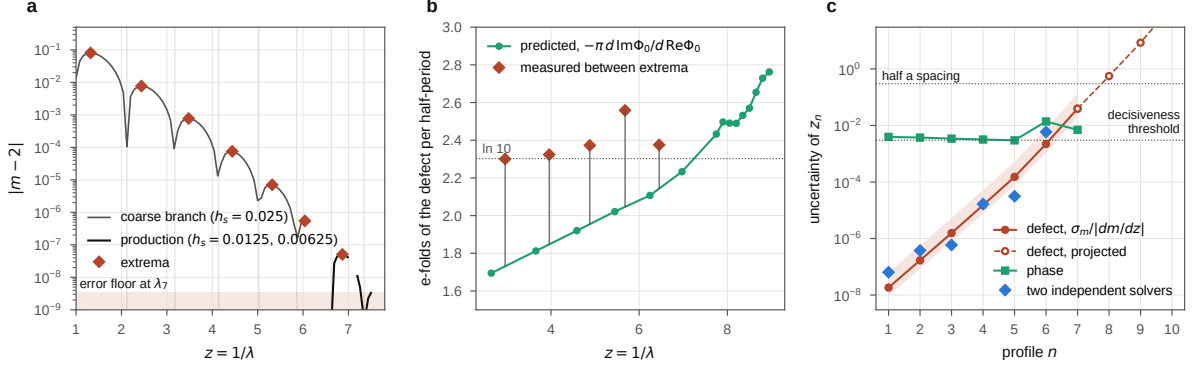


Fig. 2 | The exponential precision wall. **a**, $|m - 2|$ along the IPM branch (coarse branch, $h_s = 0.025$; production scans, $h_s = 0.0125$ and 0.00625) and its extrema, which fall tenfold per half-period, against the error floor at λ_7 . Vertical lines: the profiles. **b**, Decay of the defect per half-period predicted by the imaginary part of the phase, $-\pi d\text{Im}\Phi_0/d\text{Re}\Phi_0$, against the decay between consecutive extrema; the difference is $(0.7\text{--}1.4)\lambda$ times the prediction. **c**, Uncertainty of the profile position z_n : from the defect at a fixed floor $\sigma_m = 10^{-9}$ (band, $0.5\text{--}3.4 \times 10^{-9}$; measured crossing slopes for $n \leq 7$, projected beyond), from the phase (offset scatter and grid error divided by $d\text{Re}\Phi_0/dz$), and the difference between two independent solvers.

leaves an offset that decays roughly as $e^{s_{\text{start}}}$, from 2.1×10^{-8} at -15 to 2.2×10^{-10} at -19 (Fig. 3c). Beyond about -20 two errors grow instead: the converged Newton residual, from $0.4\text{--}2 \times 10^{-13}$ at -16 to -18 through 5×10^{-13} at -20 to 1.9×10^{-12} at -26 and 1.5×10^{-11} at -30 (Fig. 3d), and a round-off offset in m of 1.5×10^{-9} at -24 (identical at the fifth and seventh profiles, hence independent of λ), 7.7×10^{-9} at -26 and 2.1×10^{-8} at -30 , as large as the truncation error at -15 .

Together these give an erratic floor of $1\text{--}3 \times 10^{-9}$ in m at the seventh profile. The Newton floor ($\lesssim 10^{-12}$) and the far field ($s_{\text{max}} = 100 \rightarrow 160$: 6×10^{-12} , after a path-dependent 6×10^{-10} at 130) are smaller.

A pre-registered holdout the defect cannot decide

We used the deepest profile that the error budget allows as a holdout. Before any fine-grid computation below the sixth profile, we committed five predictions for $z_7 = 1/\lambda_7$ to a public repository: three rules fitted to the lower profiles (a shifted law, a geometric contraction and a three-point law; git commit c922751) and one rule from the resonance phase evaluated on a coarse branch (273ae28). The decision rule, also fixed before the crossing was known (f0b3f06), disfavours a hypothesis that misses by more than 2σ , counts the comparison as decisive only if $\sigma_z \leq 0.003$ (small enough to separate the two families of predictions, 0.007 apart, at 2σ), and takes the measured value from matched scans at two resolutions (Extended Data Table 2, Fig. 4 and Supplementary Note 3).

The committed procedure could not run as written, because its $h_s = 0.0125$ scan had been paused to test the locking error; this deviation is recorded. On the converged grid ($h_s = 0.00625$, four shifts per point) the defect gives $z_7 = 7.3451 \pm 0.0016$ (statistical). The systematic variants at the crossing move z_7 by -0.045 ($N_b = 48$), -0.137 ($N_b = 64$), -0.021 and -0.060 ($s_{\text{start}} = -22$,

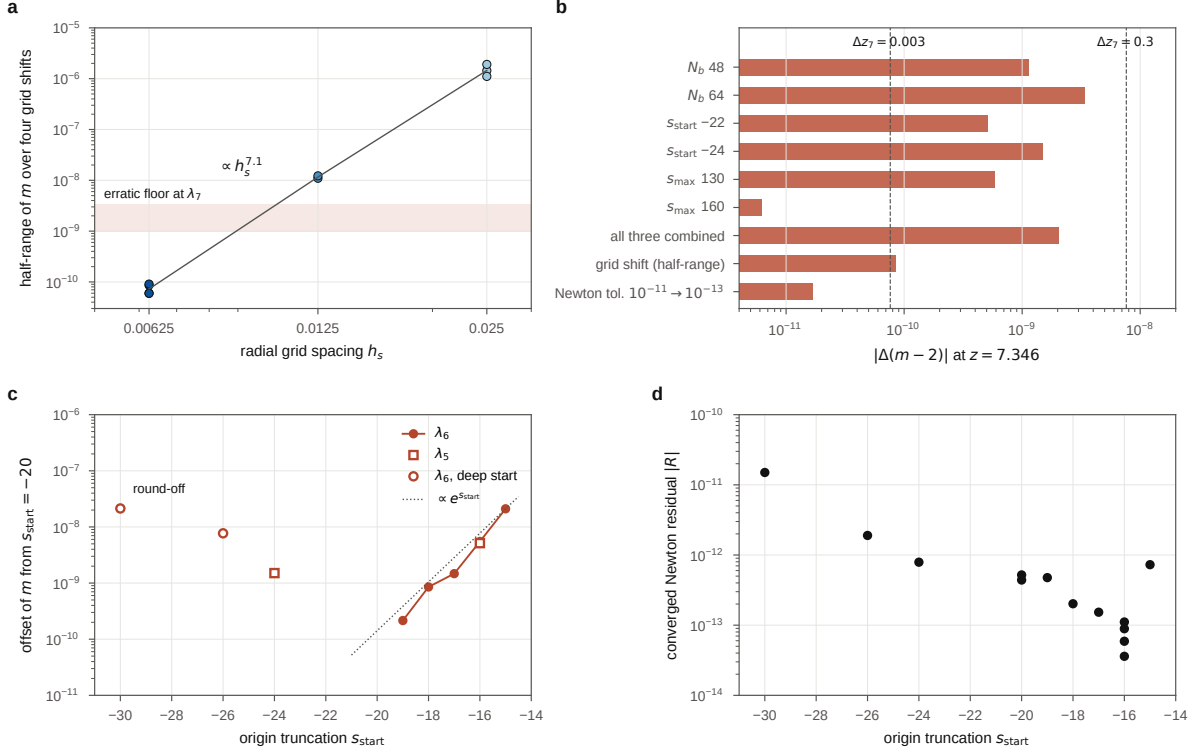


Fig. 3 | Anatomy of the error floor. **a**, Half-range of m over four grid shifts against the radial spacing h_s at $z = 7.26$ – 7.38 : front-grid locking falls as $h_s^{7.1}$. **b**, Change of m at the seventh profile under each audited parameter (production state $h_s = 0.00625$), against the changes that would move z_7 by 0.003 and 0.3. **c**, Offset of m against the origin truncation s_{start} : it decays as $e^{s_{\text{start}}}$ down to -19 ; deeper starts add a round-off offset. **d**, Converged Newton residual against s_{start} .

-24), $+0.024$ ($s_{\text{max}} = 130$) and -0.083 (all three together), so that $z_7 = 7.35 \pm 0.08$. Every registered prediction lies within 0.2σ and σ_z exceeds the decisiveness threshold about 25-fold; as the rule requires, the holdout is recorded as not decisive and no hypothesis is preferred. Five predictions spanning 0.018 in z cannot be told apart even though every solve reaches a Newton residual below 10^{-12} : the defect has run out of significant digits one profile after the sixth, where σ_z was 0.002.

A resonance phase that bypasses the wall

The hierarchy has a physical origin. As ε decreases, the self-similar flow along the wall stalls in a layer between the stagnation point and a front (Fig. 1a). Perturbations travel through this layer as short waves, $\exp(i \int \kappa ds/D_0)$, with $s = \ln x$ and $D_0 = \varepsilon/m$ the stagnation-point value of the wall speed $D = V_1/x$. The local wavenumber $\kappa(s)$ is the root of an ordinary-differential eigenproblem across the layer at each wall point (Methods), and the smooth profiles are resonances of the wave:

$$\text{Re } \Phi_0(\lambda_n) = n\pi + \delta, \quad \Phi_0 = \frac{1}{D_0} \int \kappa ds, \quad (2)$$

integrated from the stagnation point to the front cut-off $D/D_0 = 2$ (Fig. 1c and Supplementary Note 4). Three properties make Φ_0 a computational observable.

It is a quadrature over $O(1)$ data. The local problem needs only the wall data of the profile

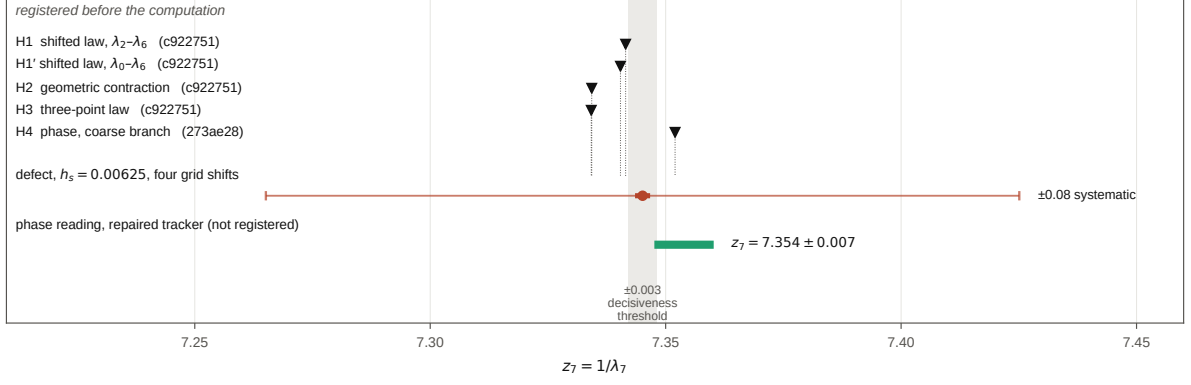


Fig. 4 | The seventh profile: a pre-registered holdout. Predictions registered before the computation (triangles; git commits in parentheses), the measurement from the defect (thick bar: statistical, ± 0.0016 ; thin bar: systematic, ± 0.08), the decisiveness threshold of the registered rule (grey band, ± 0.003) and the unregistered reading of the resonance phase (green, $z_7 = 7.354 \pm 0.007$).

— the wall speed, its strain μ and the density gradients $G = \partial_x R$ and R_y — none of which is exponentially small; Φ_0 inherits the relative accuracy of the profile, not that of the defect. At the seventh profile the inner layer ($x < 0.5$) contributes 0.886 of $I = D_0 \text{Re } \Phi_0 = 1.631$, and the receding outer structure the rest (Fig. 1b).

It has an exact scaling. In IPM the vorticity is slaved to the density gradient, $\Omega = -\partial_1 R$. Rescaling the wall-normal coordinate by $\hat{D} = D/D_0$ removes the wall speed from the local problem, so that

$$\kappa = \frac{K(G, \mu, R_y)}{\hat{D}}, \quad \Phi_0 = \int \frac{K}{D} ds. \quad (3)$$

Numerically K is invariant to 10^{-7} when $\hat{D}$ is varied from 0.012 to 2 at fixed (G, μ, R_y) (Fig. 5d). $K \approx G$ for small G and levels off at 0.8–1.05 for $G \gtrsim 3$ on the approach and the dip (Fig. 5c), so Φ_0 is close to a weighted travel time $\int ds/D$, a geometric property of the stalled layer: along the branch $I = 0.76 T + \text{const}$ for the shallow profiles and $0.81 T + \text{const}$ for the deep ones, with $T = \int ds/\hat{D}$.

It is stable where the defect is not. We recomputed Φ_0 on every variant state of the audit at the seventh profile (Fig. 5b and Extended Data Table 1). Changes that move z_7 by up to 0.137 through the defect move it by at most 0.0012 through the phase. $N_b = 32 \rightarrow 48$ shifts $\text{Re } \Phi_0$ once by 0.005, after which $N_b = 48$ and 64 agree to 10^{-4} while the defect keeps moving; s_{start} and s_{max} change $\text{Re } \Phi_0$ by less than 5×10^{-5} ($\Delta z < 10^{-5}$) against 0.02–0.06 through the defect; grid shifts change it by at most 0.003 ($\Delta z \leq 0.0006$) against up to 0.006. The phase moves 10 to more than 1,000 times less than the defect.

The offsets $\delta_n = \text{Re } \Phi_0(\lambda_n) - n\pi$ are 2.0331, 2.0227, 2.0320, 2.0360 and 2.0301 for $n = 1$ –5, constant to ± 0.005 (0.2% of a spacing) with no adjustable parameter beyond the cut-off (Fig. 5a). At $n = 6$ the offset is 1.974 ± 0.009 , a drift of -0.057 that we carry as an uncertainty. The drift is not an artefact of the cut-off: with cut-offs $D/D_0 = 1.5, 3$ and 5 the offsets trend differently over $n = 1$ –5, but each drops by an extra 0.04 ± 0.01 at $n = 6$ (Supplementary Note 4). At the seventh profile the phase reads $z_7 = 7.354 \pm 0.007$, the range spanned by $\delta = 1.974$ and 2.031; this is eleven times tighter than the defect and consistent with it. The grid dependence of $\text{Re } \Phi_0$

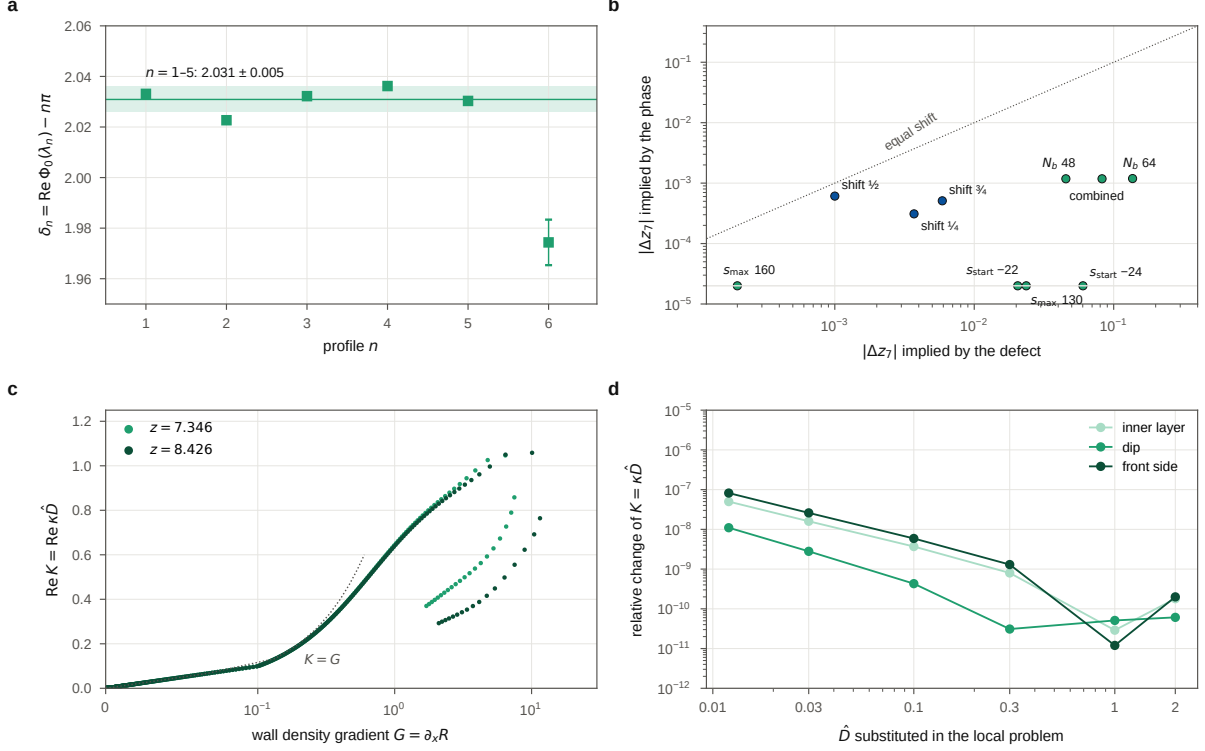


Fig. 5 | The phase is stable where the defect is not. **a**, Offsets $\delta_n = \text{Re } \Phi_0(\lambda_n) - n\pi$: 2.031 ± 0.005 for $n = 1-5$ (band) and 1.974 ± 0.009 at $n = 6$. **b**, For every audited change at the seventh profile, the shift of z_7 implied by the phase against that implied by the defect; changes below 2×10^{-5} are drawn at that value. Dotted: equal shift. **c**, $\text{Re } K = \text{Re } \kappa \hat{D}$ against the wall density gradient G along two profiles. **d**, Exact scaling: relative change of K when the local problem is solved with $\hat{D}$ replaced by 0.012–2 at fixed (G, μ, R_y) , at three wall points.

at that depth is 0.010 ($\Delta z = 0.002$). This reading is not one of the registered predictions; the registered phase rule, evaluated on a coarser branch with an earlier tracker, gave 7.352.

The imaginary part of the same quadrature predicts the wall (Fig. 2b). If the defect behaves as $|A|e^{-|\text{Im } \Phi_0|} \cos(\text{Re } \Phi_0 + \arg A)$ ^{16,17}, its envelope decays by $e^{-|\Delta \text{Im } \Phi_0|}$ per half-period. From the computed phase, $-\pi d\text{Im } \Phi_0/d\text{Re } \Phi_0 = 1.73, 1.85, 1.95, 2.05$ and 2.15 e-folds per half-period at $z = 3.0-6.5$, against measured decays of 2.30, 2.32, 2.37, 2.56 and 2.38. The deficit is $(0.7-1.4)\lambda$ times the prediction, the size of the $O(\lambda)$ corrections omitted at leading order. The precision that a direct method needs at profile n is therefore itself computable from the same $O(1)$ quadrature that locates the profile. The phase has a second, physical role that we use only as a consistency check: in 2D Boussinesq and the Hou–Luo model¹⁸ it also quantizes the unstable spectrum of the profiles, and the n -th profile has exactly n unstable directions (Supplementary Note 1).

A failure that passed every residual check

The phase requires one root $\kappa(s)$ of the local eigenproblem to be tracked along the wall, and this tracking failed in a way that residuals could not reveal. On the deep states ($z > 7.5$) the wall speed rises from $\hat{D} \approx 0.3$ at the dip to 2 at the cut-off within a few tracking steps, and κ changes by the same factor. The original tracker predicted each root from the previous κ , missed it, and

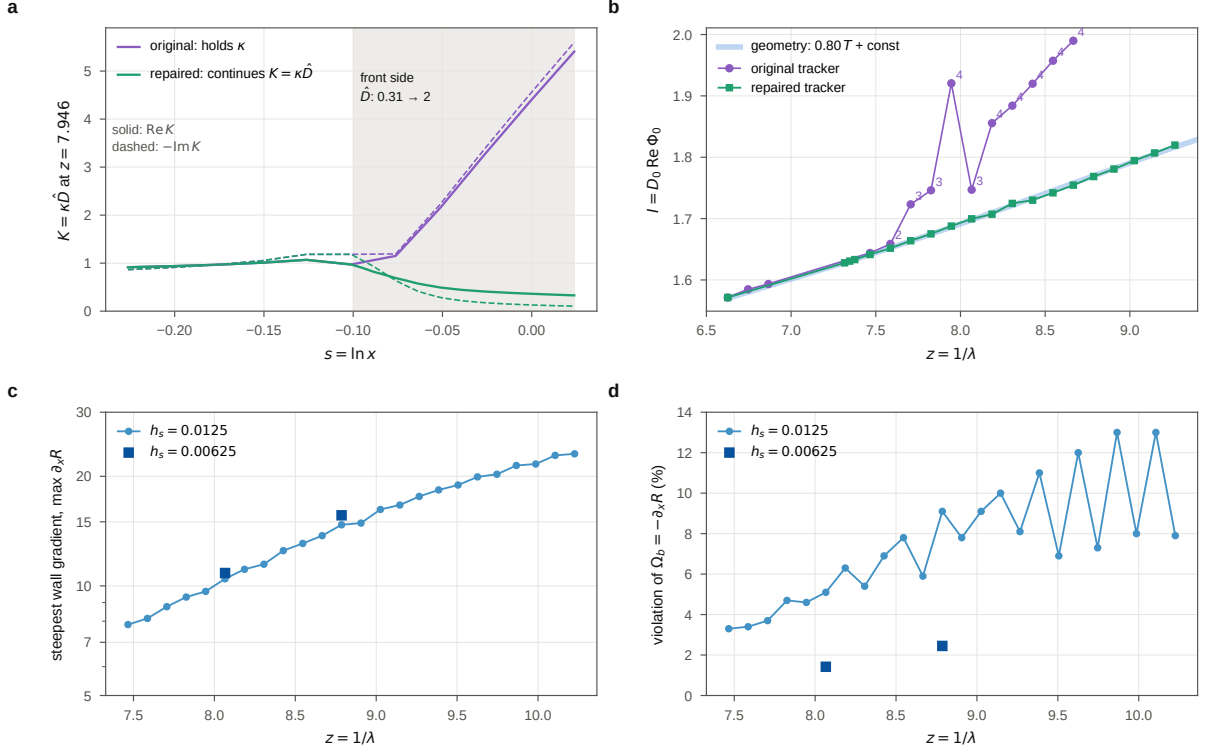


Fig. 6 | A failure that residuals missed, and the deep branch. **a**, $K = \kappa \hat{D}$ along the front side of the deep state $z = 7.946$ (shaded): the original tracker holds κ , so K grows with $\hat{D}$; the repaired tracker continues K . Solid, $\text{Re } K$; dashed, $-\text{Im } K$. **b**, $I = D_0 \text{Re } \Phi_0$ from both trackers against the geometric travel time ($0.80T + \text{const}$); numbers: points the original tracker counted as failed but carried on. **c**, Steepest density gradient on the front side along the deep branch, at $h_s = 0.0125$ and in two checks at $h_s = 0.00625$. **d**, Violation of the exact identity $\Omega_b = -\partial_x R$ (resolution indicator).

on failure held κ fixed, which inflates $K = \kappa \hat{D}$ in proportion to $\hat{D}$ (Fig. 6a): at $z = 7.946$, K reached $5.4 - 5.6i$ at the cut-off where the continued root is $0.33 - 0.10i$, and I was 14% too high — an error of 3.7 in $\text{Re } \Phi_0$, more than one profile spacing. Every value returned was a genuine root of the local problem or a held value, two roots lie close together on the front side, and every profile had a Newton residual below 10^{-12} . The tracker did count its failed points, zero on every shallow state and two to four on every deep one, but it carried on, because on the shallow states such points had been harmless.

The error was exposed by geometry. The old $I(z)$ was non-monotone (1.659, 1.723, 1.746, 1.921, 1.747, 1.856, ...) whereas the travel time T of the same profiles grows smoothly (Fig. 6b). The repair uses the exact scaling (3): beyond the dip the tracker continues K rather than κ , with predictor $K_{\text{prev}}/\hat{D}$ on a four times finer step, accepts a root only if K is continuous to 25%, holds K (not κ) at a failed point, and excludes any state with a failed point. It reproduces the shallow values ($\text{Re } \Phi_0 = 11.4568$ against 11.4566 at $n = 3$) and removes the jump; one of 17 deep states ($z = 9.386$, 11 failed points) is excluded (Supplementary Note 5). Three lessons generalize: residuals certify roots, not branches; exact invariances provide both predictors and validators; and failure counters must gate results, not only log them. The failure and its repair are part of the pre-registration record.

The deep branch and an open endpoint

Following the branch to $z = 10.23$ ($\lambda = 0.098$) on $h_s = 0.0125$ shows why the endpoint of the IPM hierarchy is beyond the present method (Fig. 6c,d, Supplementary Note 6 and Supplementary Fig. 2). The dip in the wall speed deepens ever more slowly: $\hat{D}_{\min}$ falls from 0.36 to 0.20 between $z = 7.47$ and 9.51, at 0.107 per unit z early and 0.045 over the last unit, and stays at 0.19 ± 0.01 out to $z = 10.23$, while the dip and the front recede (x_{dip} from 0.88 to 1.00). The front steepens: the largest density gradient on the wall rises from 7.8 to 23, and the violation of the exact identity $\Omega_b = -\partial_x R$, our resolution indicator, grows from 3% to 7–13% at $h_s = 0.0125$. Halving h_s at $z = 8.066$ and 8.786 leaves $\hat{D}_{\min}$, the dip and front positions and T unchanged to 0.3%, lowers the indicator from 5.1% and 9.1% to 1.4% and 2.5%, raises the largest gradient by 4–6% and $\text{Re } \Phi_0$ by 0.037 and 0.017 ($\Delta z \leq 0.007$), and reduces $m - 2$ from 2.4×10^{-8} and 3.3×10^{-7} to 1.0×10^{-9} and 2.0×10^{-9} . Between these two depths, about one profile spacing apart, the error of the defect at fixed h_s grows fourteenfold while the signal falls about fifteenfold. The geometry is resolved and the slowing of the dip is real; it is the steepening front that demands resolution, faster than a uniform grid supplies it.

Along the branch the phase coefficient I grows linearly, with slope 0.098 per unit z and a weak positive curvature ($+0.005 \pm 0.002$) over $7.47 \leq z \leq 9.27$. A saturating I , the case of a stalled layer of fixed extent as in the 2D Boussinesq and Hou–Luo hierarchies (Supplementary Fig. 3), is therefore excluded. Three continuations remain consistent with the data: I keeps growing linearly, so that profiles accumulate at $\lambda = 0$ with $\lambda_n \propto n^{-1/2}$; I diverges at a finite z_c , so that profiles accumulate at $\lambda_c = 1/z_c > 0$; or the branch terminates at a singular front with a finite phase. Pole and logarithmic fits over the full range prefer $z_c \approx 10$ –13 but with $\chi^2 \approx 35$ for 20 degrees of freedom, the deep range alone admits any $z_c \geq 9.5$, and linear extrapolations of $\hat{D}_{\min}$ and of the inverse largest gradient over the last unit vanish only near $z \approx 18$ and 13 and continue to decelerate. We therefore record the endpoint class of the IPM hierarchy as unresolved; deciding it needs a mesh that follows the front.

Within the computed range the phase still predicts profiles. Before any test we registered λ_8 , λ_9 and λ_{10} from equation (2) (git commit 251bf1e), each with a band that covers the possible drift of δ , the measured grid correction and the smoothing error, and with the precision that a decisive test ($\sigma_z \leq 0.01$) needs according to the decay rate predicted by $\text{Im } \Phi_0$: $z_8 = 7.999$ [7.970, 8.017] needs $\sigma_m \lesssim 2 \times 10^{-11}$, $z_9 = 8.628$ [8.590, 8.645] needs $\sigma_m \lesssim 1 \times 10^{-12}$ and $z_{10} = 9.201$ [9.158, 9.214] needs $\sigma_m \lesssim 6 \times 10^{-14}$, below what a double-precision solver of this kind delivers (Extended Data Table 2). The band at $n = 8$ contains the rule-based predictions registered earlier and excludes the earlier phase rule, which had been evaluated with the faulty tracker. The values of $m - 2$ that we had computed in this range lie at or below the floor; they are disclosed in the record and carry no information.

Discussion

Hierarchies of unstable self-similar solutions generated by a slow layer have smoothness defects that are beyond all orders in the layer parameter. Any method that locates them as zeros of

the defect pays a precision cost that grows exponentially with depth, and this holds for neural solvers as much as for classical ones. The residual that a solver reports does not measure this cost: our profiles have residuals of 10^{-13} , while their defects are decided at 10^{-9} .

The remedy is to change the observable rather than the solver. An asymptotic invariant of the mechanism — here a WKB phase — converts an exponentially ill-conditioned root search into a well-conditioned quadrature; its imaginary part prices the wall in advance, and its exact symmetries supply the predictors and validators that make it robust. The construction applies wherever a hierarchy arises from a stalled or slow region, for example in other boundary-driven blow-up scenarios^{19,20} and in self-similar solutions of compressible and dispersive equations^{21,22}.

Three practices that we used are cheap and transferable^{23,24}. Pre-registration with fixed decision rules²⁵ turned an inconclusive computation into a reportable negative result instead of a post-hoc fit. Measuring periodic discretization errors by grid shifting exposes a locking error that convergence studies at a fixed grid offset cannot see. Cross-checking a derived observable against an independent geometric proxy caught a failure that residuals missed.

Our study has limitations. The resonance condition is derived by formal matched asymptotics and is not proved. The offset δ drifts by 0.057 at the sixth profile, which limits the phase to about 1% of a spacing deep in the ladder. The independent global solver confirms the IPM profiles only to its own floor in the defect, and its finer grids did not converge at λ_5 and λ_6 ; the deepest IPM computations rest on one solver family with extensive internal variants. The endpoint of the IPM hierarchy is open, and the registered profiles λ_8 – λ_{10} await a solver with a front-adapted mesh and, for λ_{10} , extended precision.

Methods

Self-similar formulation. In self-similar variables $y = x/(1-t)^{1+\lambda}$ and $\tau = -\ln(1-t)$, a self-similar IPM density $\rho = (1-t)^\lambda R(y)$ satisfies $V \cdot \nabla R = \lambda R$ with $V = (1+\lambda)y + U$, $U = \nabla^\perp \Psi$ and $-\Delta \Psi = \Omega = -\partial_1 R$ on the half-plane with no penetration. By symmetry the problem is posed on the quarter plane, in log-polar coordinates (s, β) with $s = \ln r$ and $\beta = 0$ the wall. The least-singular profile is normalized at the stagnation point, where $R \simeq |y_1|^m$ with m given by equation (1) and $A = -\partial_1 U_1(0)$. The 2D Boussinesq and Hou–Luo formulations used for validation are given in Supplementary Note 7.

Marching solver. The stagnation-point structure is factored out, $R = \cos^m \beta \hat{R}$. Since all characteristics leave the origin, $\hat{R}$ is marched outwards in s from the exact local solution at $s_{\text{start}} = -20$ (implicit two-stage Gauss–Legendre for $s < 12$, classical Runge–Kutta beyond)^{26,27}. The velocity is obtained from $X = \Psi/r^2$ by eighth-order finite differences in s ²⁸ and Chebyshev collocation in β ²⁹ with $N_b = 32$, on $s \in [-120, 100]$. The fixed point $X = \mathcal{B}(\text{march}(X))$ of the march and the Biot–Savart map $\mathcal{B}$ is solved by Newton–Krylov iteration (GMRES³⁰) with central-difference Jacobian–vector products, which converges quadratically to residuals of 4 – 8×10^{-13} . The branch is followed in $z = 1/\lambda$ with secant predictors, and profiles are located by secant iteration on $m(\lambda) = 2$ and by scans on fixed z grids. One solve takes 3–8 min at

276 $h_s = 0.0125$ and 11–23 min at $h_s = 0.00625$ on one core.

277 **Independent global solver.** The unknowns are the unfactored fields (R, Ω, X) on the (s, β)
 278 grid, solved simultaneously with sixth-order finite differences in s (upwind-biased for transport)
 279 and Chebyshev collocation in β ($N_b = 24$). The Biot–Savart law is imposed as the sparse elliptic
 280 equation $(\partial_s + 2)^2 X + X_{\beta\beta} + \Omega = 0$, smoothness is imposed at $s_{\min}$ through the exact local data
 281 ($m = 2$), and λ is an eigenvalue fixed by the strain condition $A = 1 + \lambda/2$. Newton’s method uses
 282 the exact sparse Jacobian, checked against finite differences to 2×10^{-11} relative, and a sparse
 283 direct solver³¹. Each profile is solved for $s_{\max} = 8, 10$ and 12 and extrapolated geometrically
 284 in $s_{\max}$, at $h_s = 0.0125$ and $s_{\min} = -16$ (λ_1 at $h_s = 0.025$, $s_{\min} = -12$). At $h_s = 0.00625$, λ_4
 285 is unchanged to 3×10^{-8} ; for λ_5 and λ_6 Newton did not converge at that resolution within 15
 286 iterations, and those runs are logged and not used. The 2D Boussinesq version of this solver is
 287 described in Supplementary Note 1.

288 **Error audit and grid shifting.** At $z = 7.346$ the production state ($h_s = 0.00625$) was
 289 re-solved with $N_b = 48$ and 64 , $s_{\text{start}} = -22$ and -24 , $s_{\max} = 130$ and 160 , all three changes
 290 combined, and with the radial grid shifted by δh_s ($s_{\min} \rightarrow -120 + \delta h_s$, $\delta = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$). Grid shifts
 291 were also applied at $z = 7.256, 7.316$ and 7.376 and at $h_s = 0.025$ and 0.0125 . z_7 is the root of a
 292 weighted cubic through the shift means; its statistical error is the shift scatter at the crossing
 293 divided by the slope.

294 **Local eigenproblem and resonance phase.** At each wall point we solve

$$295 \quad [i\kappa(\hat{D} + \hat{c}Y) + \mu Y \partial_Y] \tilde{r} = -G \partial_Y \tilde{\psi} + i\kappa R_y \tilde{\psi}, \quad -\partial_Y^2 \tilde{\psi} + \kappa^2 \tilde{\psi} = -i\kappa \tilde{r}, \quad (4)$$

296 with $\tilde{\psi}(0) = 0$, no growing mode as $Y \rightarrow \infty$ and $\hat{c} = G$, where Y is the scaled wall-normal
 297 coordinate (derivation in Supplementary Note 4). The root κ is found by shooting (DOP853²⁶)
 298 from a regular start at $Y = 10^{-8}$ and secant iteration on the coefficient of the growing mode. It
 299 is tracked on a grid $\Delta s = 0.025$ from $s = -10$, with the Hou–Luo closed form as a fallback guess,
 300 and $\Phi_0 = D_0^{-1} \int \kappa ds$ is integrated to the first point beyond the dip where $\hat{D} = 2$, interpolated
 301 between grid points. The repaired tracker uses, beyond the dip, the step $\Delta s/4$, the predictor
 302 $K_{\text{prev}}/\hat{D}$, acceptance on 25% continuity of K and a held K at failed points; states with any
 303 failed point are excluded. One evaluation takes 30–60 s.

304 **Geometry diagnostics.** The travel time $T = \int_{-10}^{s_{\text{cut}}} ds/\hat{D}$ and the phase are partitioned into
 305 inner layer ($x < 0.5$), approach, dip core (below half depth) and front side. The resolution
 306 indicator is $\max |\Omega_b + \partial_x R| / \max \partial_x R$ over the dip-to-front window. Endpoint classes were fitted
 307 to $I(z)$ with linear, quadratic, pole and logarithmic forms (Supplementary Note 6).

308 **Pre-registration.** Every prediction was committed to the public repository before the com-
 309 putation it concerns; entries are never edited, and outcomes, deviations and corrections are
 310 appended (programs/P03_boussinesq_ladder/PREDICTIONS_IPM.md). Stage 1 (4dc82af) fixed

six structural predictions before any IPM continuation; stage 2 predicted λ_3 – λ_6 and an unstable spectrum by fixed extrapolation rules; stage 3 predicted λ_7 and λ_8 (c922751, 273ae28; decision rule f0b3f06); stage 4 predicted λ_8 – λ_{10} from the repaired phase (251bf1e) with a decision rule for future tests. The complete record and its timeline are summarized in Supplementary Note 3.

Linear stability. Instability indices were counted with two independent linearizations: the linearized march and Biot–Savart map T_μ , using real-mode parity and argument-principle counts³² of $\det(I - T_\mu)$ on right-half-plane rectangles, validated on the exact time-translation mode $\mu = 1$; and a shift-invert eigensolver on the global discretization³³. Details and counts are in Supplementary Note 1.

Reproducibility and computational cost. Every continuation and scan writes its state and log line after each step, and restarts reload the last two states for the secant predictor; fine-grid jobs run in queues with dependency waits. Container restarts during the campaign were absorbed without loss of computed states. The deep IPM campaign (λ_6 to λ_{10}) used about 13 core-hours of logged solver time on a four-core machine plus about one hour of phase and geometry evaluations; no GPU and no neural network was used. All figures are regenerated from the logged outputs by a single script. Software: Python 3.11, NumPy³⁴, SciPy³⁵, Numba³⁶ and Matplotlib³⁷.

Use of AI tools. [Draft for the authors to verify, correct and complete before submission.] An AI coding assistant ([system, version and provider]) was used under the direction of the authors to write and run the numerical code, to carry out the analyses, to produce the figures and to draft the text, between [dates]. All predictions were committed to the public record before the computations they concern, every reported number is regenerated from logged outputs by the released scripts, the profiles were reproduced with a second, independently written solver, and [describe the checks made by the authors]. The authors take full responsibility for the content of this paper.

Data availability

The converged profiles λ_0 – λ_6 and the states at $z = 7.346$, 8.066 and 8.786 (double precision, with their grids), all logged outputs from which every figure and table is generated, and the pre-registration record are available in the repository [repository URL] and will be archived at Zenodo [DOI to be assigned on acceptance]. Source data for Figs. 1–6 are provided in data/outputs of the submission package.

Code availability

All solvers, analysis scripts and the figure script are available in the repository [repository URL] under the MIT licence and will be archived at Zenodo [DOI] together with a Code Ocean capsule

for peer review.

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Author contributions

[Contributions, in the CRediT taxonomy.]

Competing interests

The authors declare no competing interests.

423 **Additional information**

424 **Supplementary information** is available for this paper.

425 **Correspondence** and requests for materials should be addressed to [corresponding author].

Extended Data

Extended Data Table 1 | Error audit at the seventh IPM profile. Production state at $z = 7.346$ ($h_s = 0.00625$, $N_b = 32$, $s_{\text{start}} = -20$, $s_{\text{max}} = 100$; $m - 2 = +3.0 \times 10^{-11}$). Shifts of z_7 are the changes of m divided by the crossing slope $|dm/dz| = 2.54 \times 10^{-8}$, and the changes of $\text{Re } \Phi_0$ divided by $d\text{Re } \Phi_0/dz = 4.54$.

Change	$\Delta(m - 2)$	Δz_7 (defect)	$\Delta \text{Re } \Phi_0$	Δz_7 (phase)
Grid shift $\frac{1}{4}$ cell	-9.2×10^{-11}	-0.004	-0.0014	-0.0003
Grid shift $\frac{1}{2}$ cell	$+2.5 \times 10^{-11}$	+0.001	-0.0028	-0.0006
Grid shift $\frac{3}{4}$ cell	-1.46×10^{-10}	-0.006	-0.0023	-0.0005
$N_b = 48$	-1.13×10^{-9}	-0.045	+0.0054	+0.0012
$N_b = 64$	-3.42×10^{-9}	-0.137	+0.0054	+0.0012
$s_{\text{start}} = -22$	-5.1×10^{-10}	-0.021	$< 5 \times 10^{-5}$	$< 10^{-5}$
$s_{\text{start}} = -24$	-1.51×10^{-9}	-0.060	$< 5 \times 10^{-5}$	$< 10^{-5}$
$s_{\text{max}} = 130$	$+5.9 \times 10^{-10}$	+0.024	$< 5 \times 10^{-5}$	$< 10^{-5}$
$s_{\text{max}} = 160$	$+6 \times 10^{-12}$	+0.0002	$< 5 \times 10^{-5}$	$< 10^{-5}$
$N_b = 48$, $s_{\text{start}} = -22$, $s_{\text{max}} = 130$	-2.06×10^{-9}	-0.083	+0.0054	+0.0012
$h_s = 0.0125 \rightarrow 0.00625$ (at $z = 7.316$)	—	—	+0.0105	-0.002

Extended Data Table 2 | Pre-registered predictions and outcomes (IPM). Git commits refer to the public repository; the record is append-only.

Stage (commit)	Prediction	Outcome
1 (4dc82af)	Accumulation as $\lambda \rightarrow 0$; linear law; index n ; lattice of unstable growth rates; rigid spectral flow; localization at the front	Accumulation clause and linear law failed (spacings contract by 0.92 per profile); index n ($n = 0-4$), lattice, flow and localization held
2	λ_3 - λ_6 from lower profiles; growth rates of the fourth unstable profile	Errors of 6%, 2.5%, 2% and 0.1% of a spacing; growth rates to 0.2-1.2%
3 (c922751, 273ae28; rule f0b3f06)	z_7 : 7.3415, 7.3404, 7.3343, 7.3342 (rules); 7.352 (phase)	Defect: 7.35 ± 0.08 , not decisive; phase reading (not registered) 7.354 ± 0.007
3	z_8 : 8.0087, 8.0063, 7.9899, 7.9917 (rules); 7.91-7.92 (phase, low confidence)	Untested (below the floor)
4 (251bf1e)	$z_8 = 7.999$ [7.970, 8.017], $z_9 = 8.628$ [8.590, 8.645], $z_{10} = 9.201$ [9.158, 9.214]; required $\sigma_m \lesssim 2 \times 10^{-11}$, 10^{-12} , 6×10^{-14}	Open benchmark